

MUHAMMAD AHMAD USMAN

Junior Full Stack Developer | MERN Stack (React, Node.js, Express, MongoDB)

Chiniot, Punjab, Pakistan | +92 327 1188084 | mahmadusman61@gmail.com | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

7th-semester BSCS student and full-stack developer who builds production-style MERN applications with real-time features, role-based access, live analytics and automated testing (169 tests in one project). Comfortable owning a project end-to-end: frontend, API, database, authentication and deployment. Adds hands-on Python/LLM experience from a Generative AI internship, including QLoRA fine-tuning and RAG. Seeking a Junior Full Stack Developer role — available immediately for part-time, remote work.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, Python, HTML5, CSS3
- **Frontend:** React, React Native (Expo), Vite, Tailwind CSS, React Router, TanStack Query, Framer Motion, Recharts
- **Backend:** Node.js, Express.js, REST API Design, JWT Authentication, Role-Based Access Control (RBAC), Socket.IO, WebRTC, Multer
- **Databases:** MongoDB, Mongoose (ODM), MongoDB Atlas
- **Testing:** Vitest, Supertest, React Testing Library, Playwright (E2E)
- **DevOps & Tools:** Docker, Git/GitHub, GitHub Actions (CI/CD), Vercel, Railway, Render, Cloudinary
- **AI / ML:** LLM Fine-Tuning (QLoRA, Unsloth), RAG, FAISS, Whisper, Ollama, Google GenAI SDK, Prompt Engineering, Pandas

PROJECTS

JobFlow — Full-Stack Job Application Management Platform | *React, Node.js, Express, MongoDB, Socket.IO, JWT, Docker*
[GitHub](#)

- Designed JWT authentication (httpOnly cookies, bcrypt hashing) with role-based access control for user and admin roles.
- Implemented full CRUD for applications, companies, interviews and notes on MongoDB/Mongoose, with server-side search, filtering, sorting and pagination.
- Delivered real-time notifications through Socket.IO private per-user rooms, alongside a live analytics dashboard driven by MongoDB aggregation pipelines.
- Integrated Cloudinary resume uploads with server-side file validation.
- Wrote 90+ automated tests (Vitest, Supertest, React Testing Library) and containerized the app with Docker; GitHub Actions CI runs lint, test and build on every push.

E-Commerce Analytics Dashboard | *React, Node.js, Express, MongoDB, Recharts*
[GitHub](#)

- Powered every revenue, order and category chart with real MongoDB aggregation pipelines — no hardcoded data.
- Implemented server-side filtering, search, pagination and CSV export, shared consistently across charts, tables and stat cards.
- Hardened the API with Helmet, rate limiting and environment-aware CORS.
- Wrote 169 automated tests (Vitest, Supertest, React Testing Library) plus Playwright E2E tests.
- Deployed frontend and backend as separate Railway services connected to a MongoDB Atlas cluster.

OurWorld — Secure Real-Time Messaging App | *React Native (Expo), Node.js, Express, MongoDB, Socket.IO, WebRTC*
[GitHub](#)

- Built a cross-platform mobile chat app with a Node.js/Express/MongoDB backend and real-time messaging over Socket.IO.
- Added end-to-end message encryption (NaCl), voice notes, WebRTC calls and Cloudinary media uploads.
- Integrated an AI companion using the Google GenAI SDK.
- Audited and fixed authentication middleware, schema and socket-handling issues across frontend and backend.

CloudNest — Full-Stack Mini Cloud Drive | *React, Node.js, Express, Multer*
[GitHub](#)

- Implemented security-conscious, disk-backed uploads: server-generated UUID filenames, resolved-path checks against path traversal, and independent server-side MIME/size validation.
- Built drag-and-drop uploads with live progress (XMLHttpRequest), nested folders and a JSON metadata store behind a REST API.
- Added search, sort/filter, favorites and a soft-delete/restore Trash system.

INTERNSHIPS & TRAINING

Web Development Intern — DecodeLabs (Remote)

Jun 2026 – Jul 2026

[Project 1](#) | [Project 2](#) | [Project 3](#) | [Project 4](#)

- Completed four front-end milestones: a W3C-valid, accessible semantic HTML/CSS page; a fluid responsive layout; a vanilla-JS DOM and state-management app (Input-Process-Output model); and a form with client-side validation.

Generative AI Intern — Arch Technologies (Remote)

Mar 2026 – Apr 2026

[RAG Pipeline](#) | [Speech-to-Reasoning](#) | [Local LLM Chatbot](#) | [Medical LLM Fine-Tuning](#)

- Built a RAG pipeline with Unsloth 4-bit quantization (Llama-3.1-8B) and FAISS for low-VRAM local document retrieval.
- Developed an end-to-end speech-to-reasoning pipeline: Whisper transcription feeding a 4-bit quantized Qwen2.5-7B model via Unsloth.
- Created a memory-enabled local chatbot with Streamlit and Ollama (Phi-3) that runs entirely offline.
- Fine-tuned a medical-domain LLM with QLoRA and Unsloth on Google Colab, adapting a general-purpose model to healthcare language on limited compute with reduced GPU memory use.

Python Programming Intern — CodeAlpha (Remote)

Feb 2026 – Mar 2026

- Completed Python automation and problem-solving tasks (data types, loops, functions, data structures) with Git/GitHub version control — see [codealpha_tasks](#).

EDUCATION

BS Computer Science — University of Agriculture, Faisalabad

- Currently in 7th semester | Expected Graduation: 2027

FSc (Pre-Medical) — Masoomen College, Chiniot

CERTIFICATIONS

Google / Coursera: Introduction to AI, Maximize Productivity With AI Tools, Crash Course on Python (2025–2026)

LANGUAGES

English (Professional) • Urdu (Native)